Direct-Actuation Bead Tracing versus RL-Tuned Pure Pursuit for Contour and Racetrack Following

Dileep Banothu
[Department]
[Institution]
[City, Country]
[email]

N. Sanjay Varma
[Department]
[Institution]
[City, Country]
[email]

***Abstract*—Pure Pursuit (PP) path trackers depend on a hand-selected lookahead distance and steering gain, and recent work delegates the scheduling of these two parameters to a reinforcement-learning (RL) policy. We ask a complementary question: can a compact policy that actuates a point-mass "bead" *directly*, without an intermediate geometric law, track references more tightly than parameter-tuned PP? We present *Bead-PPO*, a pipeline that (i) converts an arbitrary raster image into an ordered, arc-length-resampled reference polyline via Otsu segmentation and Zhang–Suen skeletonization; (ii) models a frictional, saturating point mass in a Gymnasium environment; and (iii) trains the actor of a PPO actor–critic network by expert anchoring and Dataset Aggregation (DAgger) across three F1TENTH-derived circuits. Within the same arena and scoring protocol, we re-implement RL-tuned PP together with fixed-lookahead and velocity-scheduled PP. On Hockenheim, Montreal, and Yas Marina, Bead-PPO covers 99.1%, 96.0%, and 95.7% of the reference within 0.10 m, versus 94.6%, 90.5%, and 90.1% for joint RL–PP, and lowers the tracking RMSE by 22–40% relative to the strongest PP baseline. This precision comes at a slower, damping-limited traversal. On 50 held-out synthetic contours, all three evaluated controllers exceed 96% coverage, so the advantage of direct actuation appears chiefly on geometrically demanding circuits.**

***Index Terms*—Path tracking, Pure Pursuit, imitation learning, DAgger, Proximal Policy Optimization, contour following, skeletonization, F1TENTH, autonomous racing**

I. Introduction

A path tracker turns a planned reference into actuation, and it must follow that reference faithfully across a wide range of curvatures and speeds. In scaled racing platforms such as F1TENTH [1], geometric trackers, most prominently Pure Pursuit (PP) [2], [3], dominate because they are analytically tractable and cheap enough to run at high rates on embedded hardware. PP steers toward a target point one *lookahead distance* L_d ahead on the reference, so its behavior is governed by this scalar and by an effective steering gain g. A short lookahead yields agile but oscillatory motion on straights; a long one yields smooth motion that understeers in hairpins. Heuristic schedules that tie L_d to speed only partially resolve this tension.

One response is to learn the schedule. Elgouhary and El-Wakeel [4] keep the PP law intact and let a PPO [5] policy select (L_d, g) online from speed and curvature-preview features, which preserves interpretability and adds only two scalar interfaces to an existing control stack. The opposite response is to drop the geometric law altogether and let a learned policy command the actuation directly. The two designs are rarely compared on equal terms, in one simulator with one set of metrics.

This paper provides such a comparison. We had independently built a *bead-tracing* simulator in which a point mass follows a contour extracted from an arbitrary image. Inside this arena we re-implement the RL–PP environment, reward, and training recipe of [4] and ask one question: when the references and the metrics are the same, does a directly actuated policy track more precisely than a PP law whose parameters are learned?

The answer is qualified. On three F1TENTH-derived circuits, the direct-actuation policy covers more of the reference and has lower tracking error than every PP variant, but it moves at less than half their pace and was trained on those circuits. On short held-out contours, a hand-tuned PP is as accurate or slightly better. Our re-implementation of RL-tuned PP also does not reproduce the benefit of learning the steering gain reported in [4]. Specifically, we contribute:

- an image-to-reference front end that recovers an ordered, uniformly resampled polyline from skeletons or contours, so that arbitrary drawings and racetrack maps become tracking tasks without manual annotation;
- *Bead-PPO*, a direct-actuation tracker whose PPO actor–critic network has its actor trained by expert anchoring and multi-circuit DAgger [6];
- an arena-scaled re-implementation of RL-tuned PP [4], with its joint and lookahead-only variants and two classical PP baselines, all scored under one protocol;
- a circuit-level and zero-shot comparison, an analysis of the learned (L_d, g) schedule, and an explicit account of the precision–pace trade-off and of threats to validity.

II. Related Work

Geometric tracking. Pure Pursuit [2] and the Stanley controller [7] are the canonical geometric trackers; Snider [3] offers a comparative treatment of both, and kinematic models suitable for their design are discussed in [8], [9]. The sensitivity of PP to its lookahead has motivated numerous adaptive schemes that scale L_d with speed or curvature, yet these rules

retain fixed functional forms whose coefficients seldom transfer across tracks. Minimum-curvature raceline optimization [10] typically furnishes the reference such trackers follow in racing [11].

Learning-based control. Deep RL has been applied extensively to driving [12] and to robotics generally [13]. PPO [5], with Generalized Advantage Estimation (GAE) [14], is a de facto on-policy baseline whose performance is known to depend heavily on implementation details [15]. Off-policy alternatives such as Soft Actor-Critic (SAC) [16] trade sample efficiency against stability. End-to-end policies can be highly expressive but often raise interpretability and transfer concerns; comparative studies in racing [17] document these trade-offs. Hybrid approaches that tune classical controllers with RL, such as the RL-tuned PP of [4], occupy an intermediate position.

Imitation learning. Behavior cloning dates to ALVINN [18] but suffers from compounding error once the learner drifts from the expert's state distribution. DAgger [6] remedies this by iteratively labeling the states that the learner itself visits and aggregating them into a growing dataset, with no-regret guarantees. We use this property to train the actor of a PPO actor–critic network by supervised updates while leaving its critic unchanged.

Positioning. Our tracker sits between these lines of work. Unlike RL-tuned PP [4], it is not constrained to a PP steering law; unlike purely end-to-end RL, it relies on a cheap geometric expert to remove the exploratory burden on long, thin contours.

III. Framework and Methodology

Fig. 1 summarizes the pipeline. A perception front end turns an image into a reference polyline (Section III-A); a point-mass "bead" is then trained to follow it (Sections III-B–III-E); and the RL–PP baseline is re-implemented on the same references (Section III-F).

A. Raster-to-Reference Perception

An input image is resized so that its longer side does not exceed 1200 px and converted to grayscale. After Gaussian blurring, a foreground mask is obtained with Otsu's method [19]; the threshold polarity is chosen from the median intensity of the image border so that both dark-on-light and light-on-dark drawings are admissible. A 3×3 morphological closing removes pinholes. If the foreground occupies at most 20% of the image, the stroke is thin enough to be *skeletonized* with the Zhang–Suen thinning algorithm [20]; otherwise, closed external contours are extracted directly. Each 8-connected skeleton component is ordered into a polyline by a greedy walk that begins at an endpoint and, at every junction, prefers the neighbor that minimizes the turning angle. The resulting components are mapped to a 20×20 m arena with the y-axis inverted to a right-handed frame, and components shorter than 0.1 m are discarded. The remainder are re-sampled at uniform arc length, with a total budget of approximately 350 waypoints distributed in proportion to component length. Indices at which disjoint components meet are recorded as *path breaks* and excluded from all segment-distance computations.

B. Bead Dynamics

The bead is a point mass of radius 0.15 m with position $\mathbf{p}_t \in \mathbb{R}^2$ and velocity $\mathbf{v}_t$. A normalized action $\mathbf{u}_t$, with each component in $[-1, 1]$, commands an acceleration of up to $a_{\max} = 10$ m/s^2 per axis. Viscous friction $\mu = 7$ s^{-1} and a speed ceiling $v_{\max} = 7$ m/s act on the motion, which is integrated with $\Delta t = 0.016$ s:

$$\begin{aligned} \mathbf{v}_{t+1} &= (\mathbf{v}_t + a_{\max}\mathbf{u}_t\Delta t)(1 - \mu\Delta t), \\ \mathbf{p}_{t+1} &= \mathbf{p}_t + \mathbf{v}_{t+1}\Delta t, \end{aligned} \tag{1}$$

with speed clipping at $v_{\max}$ and an inelastic reflection (restitution 0.15) at the arena wall. Under a sustained unit action along one coordinate axis, (1) converges to the terminal speed

$$v^* = a_{\max}\Delta t\,(1 - \mu\Delta t)/(\mu\Delta t) \approx 1.27 \text{ m/s}, \tag{2}$$

and a diagonal command with both components saturated reaches at most $\sqrt{2}\,v^* \approx 1.79$ m/s. The damping, not the nominal ceiling, therefore bounds the achievable pace. This quantity is central to interpreting the traversal times in Section V.

C. Observation and Action Spaces

The observation $\mathbf{o}_t$, a 16-dimensional vector with components in $[-1, 1]$, is egocentric and scale-normalized by the arena half-width $N = 10$ m. It comprises the offsets to the current and next waypoints (4), the normalized position (2) and velocity (2), the distance to the nearest reference point within a $\pm$20-index window (1), an on-curve indicator (1), the heading direction cosines (2), a discrete signed turning-angle proxy for curvature at the nearest point (1), the unit vector toward a five-waypoint preview (2), and the normalized progress index (1). The action is the two-dimensional acceleration command $\mathbf{u}_t$. The sequential waypoint pointer w_t advances only to indices within $+4$ of its current value that lie inside the coverage radius $\rho = 0.18$ m, which precludes short-cutting across self-approaching segments.

D. Reward Design

Let e_t denote the distance to the nearest reference point, d_t the distance to waypoint w_t, n_t the number of newly passed waypoints, and $\hat{\mathbf{r}}_t$ the local unit tangent. The per-step reward is

$$\begin{aligned} r_t = {} & w_a \operatorname{clip}\big((d_{t-1} - d_t)/\rho, -1, 1\big) + w_w n_t \\ & + w_e \exp\big[-(e_t/\sigma)^2\big] \\ & + w_\tau \operatorname{clip}\big(\mathbf{v}_t \cdot \hat{\mathbf{r}}_t / v_{\max}, -1, 1\big) - w_\ell \min(e_t/\rho_s, 3) \\ & - w_o \mathbf{1}[e_t > \rho] - w_b \mathbf{1}_{\text{back}} - w_c \mathbf{1}_{\text{wall}} \\ & - w_j \min(\|\mathbf{u}_t - \mathbf{u}_{t-1}\|, 1) - w_s, \end{aligned} \tag{3}$$

where $\mathbf{1}_{\text{back}}$ and $\mathbf{1}_{\text{wall}}$ indicate a backtracking step and wall contact, respectively, $\sigma = 0.08$ m sets the width of a Gaussian *centering* bonus, and $\rho_s = 0.10$ m is the strict tolerance. A terminal bonus of $+50$ is added when sequential coverage

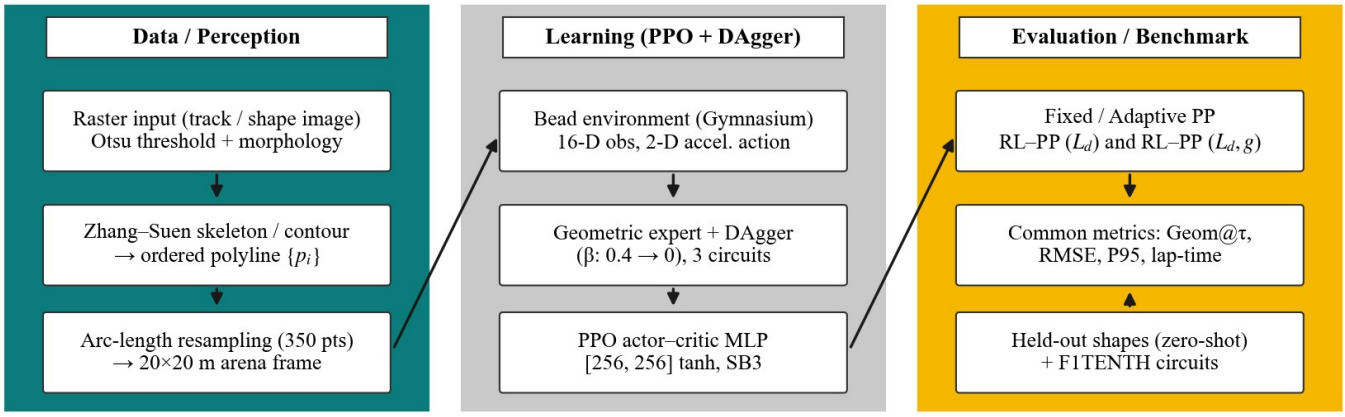


Fig. 1. Overall pipeline. A raster image is converted into an ordered reference polyline (left), on which the Bead-PPO actor is trained with expert anchoring and DAgger (center). The resulting policy and four Pure Pursuit variants—including a re-implementation of RL-tuned PP [4]—are scored with identical metrics on F1TENTH-derived circuits and held-out synthetic shapes (right).

TABLE I
REWARD WEIGHTS OF THE BEAD ENVIRONMENT

Term	Symbol	Weight	Term	Symbol	Weight
Approach	w_a	0.10	Off-curve	w_o	0.10
Waypoint	w_w	0.40	Backtrack	w_b	0.05
Centering	w_e	0.25	Wall contact	w_c	0.50
Tangency	w_τ	0.15	Action jerk	w_j	0.005
Tracking error	w_ℓ	0.05	Step cost	w_s	0.001

reaches 99%, $+15$ when an episode is truncated above 80%, and -5 otherwise. Episodes are truncated after 3500 steps or 400 consecutive steps without progress. Table I lists the weights. The Gaussian term is deliberately sharper than the linear error penalty: the latter prevents gross divergence, while the former creates a steep incentive to hug the centerline once the bead is already close. This reward defines the task and is the objective for reward-based fine-tuning; the Bead-PPO policy evaluated in this paper is trained by imitation instead (Section III-E) and does not use it as a training signal.

E. Policy Architecture and Learning

The policy is a Stable-Baselines3 [21] PPO actor–critic with separate [256, 256] tanh multilayer perceptrons for the actor and critic (Table II). Pure on-policy exploration proved inefficient on long, thin contours, where a single lateral excursion forfeits the rest of the episode. We therefore *anchor* the actor with a geometric expert that points the bead toward the next waypoint,

$$\mathbf{u}_t^* = (\mathbf{p}[w_t + 1] - \mathbf{p}_t)/\|\mathbf{p}[w_t + 1] - \mathbf{p}_t\|, \qquad (4)$$

and refine it with DAgger [6]: the learner acts, the expert labels the visited states, and the aggregated dataset is regressed with a mean-squared-error loss on the actor pathway (feature extractor, policy MLP, and action head). Algorithm 1 gives the procedure. Rollouts from all three circuits are aggregated so that a single network serves every map, and training stops once the worst-circuit coverage at 0.10 m reaches 95%, or after eight rounds.

Algorithm 1 Multi-circuit expert anchoring and DAgger refinement

Input: circuits T, expert π^* (4), PPO network π_θ, mixing schedule $\beta = (0.4, 0.2, 0.1, 0.05, 0, 0, 0, 0)$
$D \leftarrow$ 8 expert episodes per circuit; fit actor on D (12 epochs, lr $2{\times}10^{-4}$)
for round $k = 1, \ldots, 8$ **do**
  for each circuit τ in T, roll out 4 episodes executing π^* w.p. β_k, else π_θ; label every state with $\pi^*(\mathbf{o})$
  $D \leftarrow D + \{(\mathbf{o}, \pi^*(\mathbf{o}))\}$; refit actor (8 epochs, lr $1.5{\times}10^{-4}$ for $k \le 3$, else $7{\times}10^{-5}$)
  evaluate Geom@0.10 on every τ; checkpoint if the minimum improves
  if $\min_\tau$ Geom@0.10 ≥ 0.95 **then break**
return best checkpoint

Two points about this procedure matter for interpreting the results. First, no PPO clipped-surrogate updates are applied to the evaluated policy: the critic is left untouched, and at deployment the actor mean is executed deterministically. The name Bead-PPO refers to the network parameterization, and PPO fine-tuning is left to future work. Second, the stopping and checkpoint rule is evaluated on the same circuits used in Table III, so those circuits are in-distribution for Bead-PPO (Section V-F).

F. Re-implementation of RL-Tuned Pure Pursuit

The baseline reproduces the formulation of [4] inside the same arena, so that both controllers see identical references and scoring. The vehicle follows the kinematic bicycle model [8] with wheelbase $W = 0.33$ m. The PP law converts the body-frame lateral offset y' of the lookahead point into a

TABLE II
LEARNING HYPERPARAMETERS

Quantity	Bead-PPO	RL–PP (re-impl. of [4])
Policy / value MLP	[256, 256] tanh	SB3 default [64, 64]
Rollout n_{steps} / batch	2048 / 128‡	2048 / 256
Epochs per update	10‡	5
γ / GAE λ	0.99 / 0.95‡	0.99 / 0.98
Clip ε / target KL	0.2 / —‡	0.2 / 0.015
Entropy / value coeff.	0.001 / 0.5‡	0.02 / 0.6
Grad-norm clip	0.5	0.7
Learning rate	see Alg. 1	2.4×10^{-4}, linear decay
Normalization	—	VecNormalize (obs, return)
Actor training	expert + DAgger	PPO with reward teacher terms

‡ PPO settings of the network configuration; not exercised by the DAgger training (Alg. 1) that produced the reported Bead-PPO policy.

curvature command, which is mapped to a steering angle:

$$\gamma = g \cdot 2y'/L_d^2, \quad \delta = \text{clip}(\arctan(W\gamma), \pm 1.10 \text{ rad}),$$
$$d\theta/dt = (v/W)\tan\delta. \tag{5}$$

Following [4], the policy observes speed, the absolute curvatures $\kappa_{0,1,2}$ at waypoint offsets $\{0, 5, 12\}$, and $\Delta\kappa = \kappa_1 - \kappa_0$. We append the signed lateral error and $(\sin e_\theta, \cos e_\theta)$ of the heading error, giving an 8-D input. The action is decoded to L_d in $[0.20, 1.20]$ m and g in $[0.60, 1.40]$ and passed through the first-order smoother of [4] with $\beta_L = \beta_g = 0.2$:

$$\tilde{L}_{t+1} = \beta_L L_{t+1} + (1-\beta_L)\tilde{L}_t, \quad \tilde{g}_{t+1} = \beta_g g_{t+1} + (1-\beta_g)\tilde{g}_t. \tag{6}$$

Because the arena is several times smaller than a full-scale F1TENTH circuit, the published ranges and teacher coefficients were rescaled. The teacher targets become $L^* = \text{clip}(0.65 - 0.18\kappa^{\max} + 0.04v, 0.26, 0.85)$ and $g^* = \text{clip}(0.96 + 0.14\kappa^{\max}, [g_{\min}], [g_{\max}])$. The longitudinal reference is a friction-limited profile $v_{\text{ref}} = \text{clip}(4.2/\sqrt{1 + 2.5|\kappa|}, 1.8, 5.0)$ m/s, followed by backward (braking, 3.5 m/s^2) and forward (acceleration, 2.5 m/s^2) passes and tracked with $a = 3(v_{\text{ref}} - v)$. The reward keeps the structure of [4]—teacher regularization, parameter-jerk penalties, a curvature-dependent lookahead incentive, progress, speed, collision, and stall terms, clipped to $[-30, 100]$—and augments it with a Gaussian corridor term $3.5\exp[-\frac{1}{2}(e/0.035)^2]$, a quadratic penalty beyond 0.07 m, and a heading-alignment bonus $0.8\cos e_\theta$.

Four PP controllers are derived from this environment: *Fixed PP* ($L_d = 0.55$ m, $g = 1.05$); *Adaptive PP* with the velocity-linear schedule $L_d = \text{clip}(0.35 + 0.25v, [L_{\min}], [L_{\max}])$; *RL–PP (joint)*, which executes the learned (L_d, g); and *RL–PP (L_d only)*, which uses the lookahead channel of the same trained network while holding $g = 1.05$.

IV. EXPERIMENTAL SETUP

A. References and Datasets

Three F1TENTH circuits—Hockenheim, Montreal, and Yas Marina—were rasterized from their public centerline files and passed through the front end of Section III-A (Fig. 2(a)–(c)). They yield closed references with sharp chicanes and hairpins whose discrete curvature reaches the clipping limit of 5 m^{-1}. Bead-PPO was trained on all three. The RL–PP policy was trained with the recipe of Table II on a single hand-drawn closed contour and is therefore zero-shot on every circuit. To probe generalization further, we generated two synthetic datasets of 1000 images each (70/15/15 train/validation/test split). The *closed-shape* set contains polygons, ellipses, and rounded quadrilaterals; the *curve* set contains open arcs, lines, and parabolas (Fig. 2(d)–(g)). The first 25 images of each *test* split, never used for training, form the held-out benchmark.

B. Metrics

Let P be the reference polyline and X the executed trajectory. *Geometric coverage* at tolerance τ is the fraction of reference points whose distance to X does not exceed τ:

$$\text{Geom@}\tau = (1/|P|)\sum_{\mathbf{p}\in P} \mathbf{1}[\text{dist}(\mathbf{p}, X) \le \tau]. \tag{7}$$

Conversely, the lateral error $E = \text{dist}(\mathbf{x}, P)$ of each trajectory sample yields the tracking RMSE, the 95th-percentile error (P95), and the maximum error. *Sequential coverage* is the fraction of waypoints passed in order, and *traversal time* is the number of steps multiplied by Δt. Geom@τ penalizes skipped geometry, which RMSE alone cannot detect; RMSE penalizes wandering, which coverage alone cannot detect.

C. Protocol

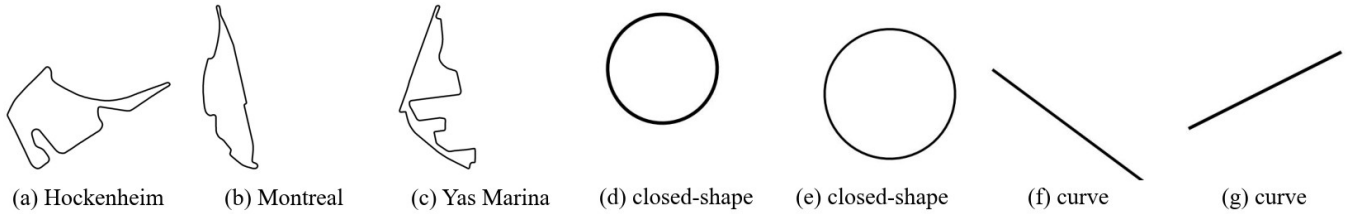
Every controller was executed for five episodes per circuit with seeds $\{50000 + 137k\}$, $k = 0, \ldots, 4$, which perturb the initial position by up to ± 5 cm. The policies were run deterministically, and each learned controller is a single trained model. All experiments ran on a CPU-only workstation using PyTorch 2.11, Stable-Baselines3 2.8 [21], and Gymnasium 1.2 [22].

V. RESULTS AND DISCUSSION

We first compare tracking accuracy on the three circuits, then examine what that accuracy costs in pace (Section V-B) and whether it transfers to unseen contours (Section V-C). We then analyze why learned PP tuning did not help in our arena (Sections V-D and V-E) and close with threats to validity.

A. Circuit-Level Tracking Accuracy

Table III consolidates the circuit results. Bead-PPO attains the highest Geom@0.10 on all three circuits (99.1%, 96.0%, and 95.7%) and the lowest RMSE, P95, and maximum error throughout; the only entry in which it is not best is Geom@0.05 on Yas Marina, where Fixed PP is marginally higher (79.8% vs. 79.1%). Across the five start perturbations, the coverage entries vary by less than 0.3 percentage points. Relative to the strongest PP variant on each circuit, which is



(a) Hockenheim (b) Montreal (c) Yas Marina (d) closed-shape (e) closed-shape (f) curve (g) curve

Fig. 2. Reference sources. (a)–(c) F1TENTH circuits used for Bead-PPO training and for all circuit-level evaluation. (d)–(g) Representative samples from the synthetic closed-shape and curve datasets, whose held-out test splits form the zero-shot benchmark.

TABLE III
TRACKING PERFORMANCE ON THE THREE CIRCUITS (MEAN OF FIVE SEEDED EPISODES). BEST PER CIRCUIT IN BOLD; TRAVERSAL TIME AND ACTION CHANGE ARE NOT RANKED.

Controller	Geom@0.05 [%] ↑	Geom@0.10 [%] ↑	RMSE [cm] ↓	P95 [cm] ↓	Max [cm] ↓	Sequential [%] ↑	Traversal time [s]	Mean $\|\Delta$ action$\|$
Hockenheim (in-distribution for Bead-PPO)								
Bead-PPO (ours)	**91.5**	**99.1**	**2.46**	**5.49**	**8.51**	99.1	42.36	0.031†
RL–PP (joint L_d, g) [4]	75.5	94.6	5.17	10.19	22.83	**99.9**	17.50	0.011
RL–PP (L_d only)	76.2	95.3	4.79	8.93	20.17	99.6	17.32	0.011
Adaptive PP ($v \to L_d$)	50.5	69.5	13.76	28.47	59.99	99.7	16.35	0.008
Fixed PP	84.9	95.9	3.62	6.88	18.22	99.0	17.37	0.013
Montreal (in-distribution for Bead-PPO)								
Bead-PPO (ours)	**86.1**	**96.0**	**3.46**	**8.03**	**13.73**	99.1	30.74	0.027†
RL–PP (joint L_d, g) [4]	75.4	90.5	6.42	14.91	29.31	**100.0**	13.16	0.013
RL–PP (L_d only)	76.6	90.5	6.02	14.41	26.39	**100.0**	13.10	0.014
Adaptive PP ($v \to L_d$)	59.1	72.0	17.76	39.34	81.38	**100.0**	12.55	0.007
Fixed PP	82.5	90.8	5.74	14.03	27.47	**100.0**	13.16	0.016
Yas Marina (in-distribution for Bead-PPO)								
Bead-PPO (ours)	79.1	**95.7**	**4.06**	**8.92**	**17.58**	99.1	41.97	0.030†
RL–PP (joint L_d, g) [4]	71.3	90.1	6.24	13.54	26.56	99.7	18.31	0.015
RL–PP (L_d only)	73.5	89.3	5.85	13.28	23.90	99.7	18.19	0.016
Adaptive PP ($v \to L_d$)	47.9	66.9	11.64	26.04	37.31	99.1	16.97	0.009
Fixed PP	**79.8**	90.6	5.22	12.16	23.22	**100.0**	18.26	0.017

† Norm of the change in the 2-D acceleration command, which is not commensurate with the steering-angle change (rad) reported for the PP variants.

always Fixed PP, the RMSE of Bead-PPO is lower by 32% on Hockenheim, 40% on Montreal, and 22% on Yas Marina (Fig. 3); relative to joint RL–PP, the reductions are 52%, 46%, and 35%.

The gain is not confined to the average. The maximum error falls by 24–53% relative to the best PP variant, and the cumulative error distributions in Fig. 4 show why: the Bead-PPO curve reaches unity at 9, 14, and 18 cm on the three circuits, whereas every PP variant has a tail extending to 18–81 cm. Fig. 5 shows the same effect along one lap of Yas Marina, where joint RL–PP leaves the tolerance band roughly three times as often as Bead-PPO.

Where Bead-PPO does fail is equally informative. The reference points it leaves beyond 0.10 m cluster at a few hairpins (Fig. 6), which Fig. 7 magnifies. On the Montreal hairpin it swings wide of the corner exit by up to about 0.14 m before re-converging, and it enters the Yas Marina chicane slightly early. At the same corners, the velocity-scheduled Adaptive PP inherits a lookahead exceeding 1 m at cruising speed and cuts the apex decisively, which accounts for its sub-75% coverage, while the other three PP variants follow nearly coincident paths. The advantage of Bead-PPO is therefore earned on the long, moderately curved stretches, where its trajectory is almost indistinguishable from the reference, rather than by superior cornering.

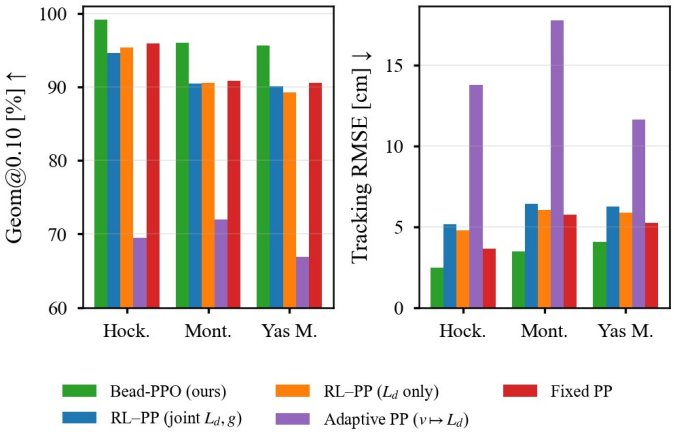


Fig. 3. Geom@0.10 (left, higher is better) and tracking RMSE (right, lower is better) by circuit and controller.

B. The Precision–Pace Trade-off

The accuracy of Bead-PPO comes at a price: it needs 2.3–2.6× the traversal time of the PP controllers (Table III). The plant, not the policy, sets this gap: by (2), viscous damping caps the bead's steady speed at about 1.3 m/s along an axis (at most 1.8 m/s diagonally), whereas the PP vehicle tracks a curvature-aware profile reaching 5 m/s. The two agents therefore operate in different dynamical regimes, and slower

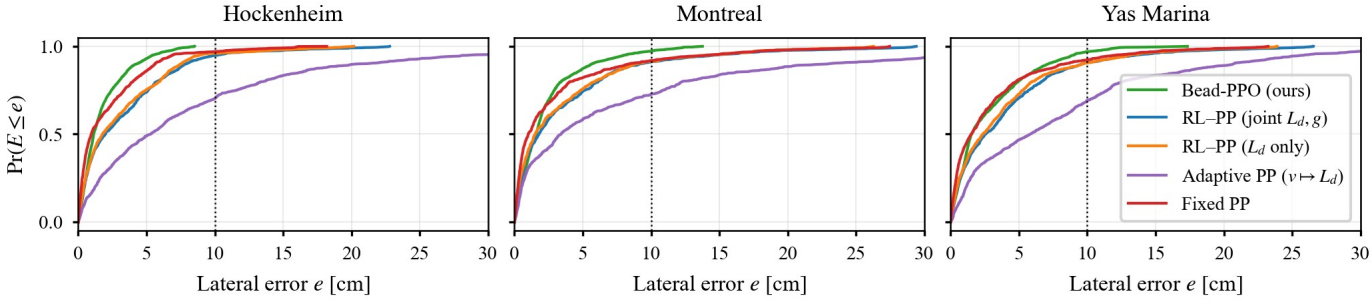


Fig. 4. Empirical cumulative distribution of the lateral error on each circuit (first seed). The dotted line marks the 0.10 m tolerance.

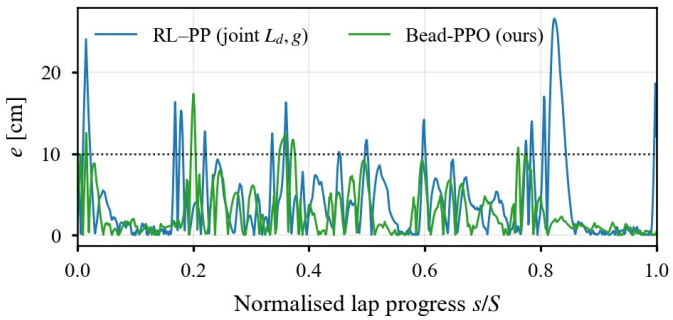


Fig. 5. Lateral error along one lap of Yas Marina. Bead-PPO (green) exceeds the 0.10 m tolerance (dotted) in four brief excursions, peaking at 17.6 cm; joint RL–PP (blue) exceeds it at roughly a dozen high-curvature segments, peaking at 26.6 cm.

TABLE IV
ZERO-SHOT RESULTS ON HELD-OUT TEST CONTOURS (25 PER SET)

Set / Controller	Geom@0.10 [%] ↑	RMSE [cm] ↓	Seq. [%] ↑
Closed shapes			
Bead-PPO (ours)	96.2 ± 3.3	3.75	**99.2**
RL–PP (joint)	97.0 ± 1.6	2.69	98.7
Fixed PP	**98.0 ± 0.8**	**2.34**	98.6
Open curves			
Bead-PPO (ours)	98.4 ± 0.7	2.71	**99.1**
RL–PP (joint)	98.2 ± 1.1	1.90	98.5
Fixed PP	**98.7 ± 0.6**	**1.57**	98.3

±: [standard deviation across the 25 shapes?]

motion inherently eases tracking. No claim about lap time can be made without a speed-matched comparison, either by raising $a_{\max}$ or by imposing the same v_{ref} on the bead. The present evidence supports a narrower conclusion: *at its native pace, direct actuation yields higher Geom@0.10 and lower RMSE, P95, and maximum error than every evaluated PP variant at its native pace.*

C. Zero-Shot Generalization to Held-Out Contours

The circuits favor Bead-PPO, which was trained on them. The 50 held-out shapes in Table IV, which no policy saw during training, are a fairer test, and here the ordering changes. All three controllers exceed 96% mean coverage, and Fixed PP, whose parameters suit curvatures of this kind, is marginally the most accurate. Bead-PPO is close in coverage on open curves (98.4% vs. 98.7%) but has the highest RMSE there (2.71 vs. 1.57 cm); it trails on closed shapes (96.2% vs. 98.0%) and exhibits the largest dispersion (±3.3 pp). This dispersion stems chiefly from sharp polygon vertices, where its constant-magnitude expert anchoring encourages it to round the corner (Fig. 8). The benefit of direct actuation is therefore concentrated on long, heterogeneous references; on short, simple contours, a well-chosen geometric law already approaches the ceiling of the metric.

D. Interpretability of the Learned PP Schedule

A further result of Table III is that neither learned PP variant beats Fixed PP. To see why, Fig. 9 dissects the joint RL–PP policy on Montreal. The learned lookahead ranges from 0.47 to 0.77 m. It is negatively correlated with previewed curvature ($r = -0.67$) and positively correlated with speed ($r = +0.56$), which qualitatively reproduces the behavior reported in [4]: the preview contracts ahead of bends and dilates on straights. The steering gain, by contrast, remains confined to a narrow band ($0.96 \le g \le 1.04$, standard deviation 0.011) and dips slightly in corners ($r = -0.64$ with $\kappa^{\max}$). This mild modulation is also consistent with [4], where g "varies more mildly" than L_d.

E. Ablation: Joint versus Lookahead-Only Tuning

Removing the nearly constant gain channel accordingly costs nothing. Unlike the results reported in [4], RL–PP (L_d only) matches or slightly improves on the joint variant in RMSE on every circuit (4.79 vs. 5.17, 6.02 vs. 6.42, and 5.85 vs. 6.24 cm). Two caveats temper this negative result. First, our L_d-only controller reuses the lookahead output of the jointly trained network rather than a separately trained policy, so it is an *inference-time* ablation, not the training-time ablation of [4]. Second, the learned g settles slightly *below* the fixed value of 1.05 used by the other controllers, reducing corrective authority exactly where it is needed. The result therefore indicates that, at arena scale and with our budget, the gain channel learned a nearly constant, slightly conservative value. It does not refute the hardware findings of [4].

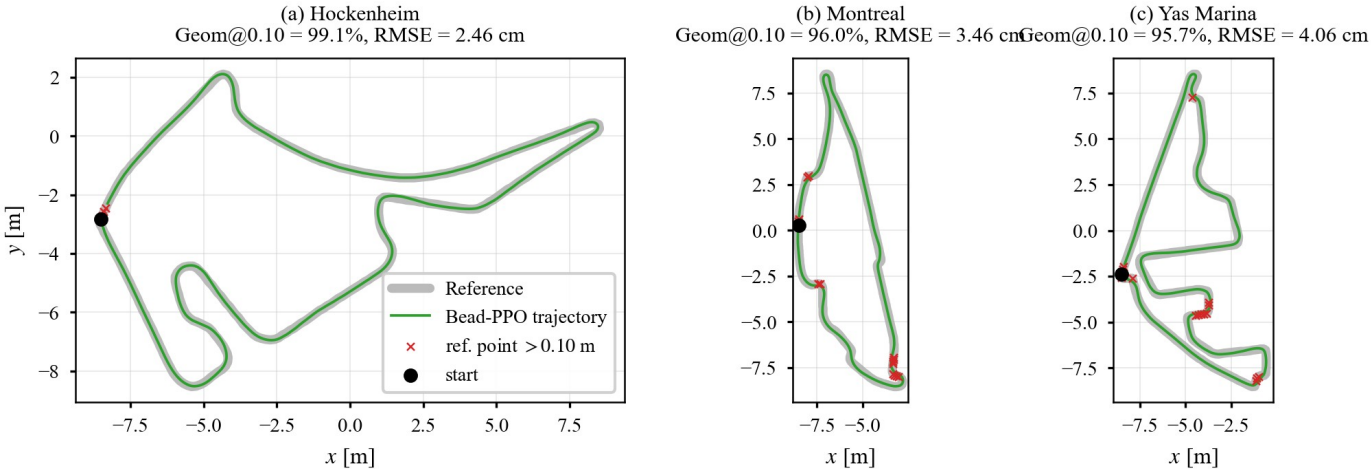


Fig. 6. Bead-PPO trajectories (green) over the three reference circuits (grey): (a) Hockenheim, Geom@0.10 = 99.1%, RMSE = 2.46 cm; (b) Montreal, 96.0%, 3.46 cm; (c) Yas Marina, 95.7%, 4.06 cm. Red crosses mark reference points left farther than 0.10 m from the trajectory.

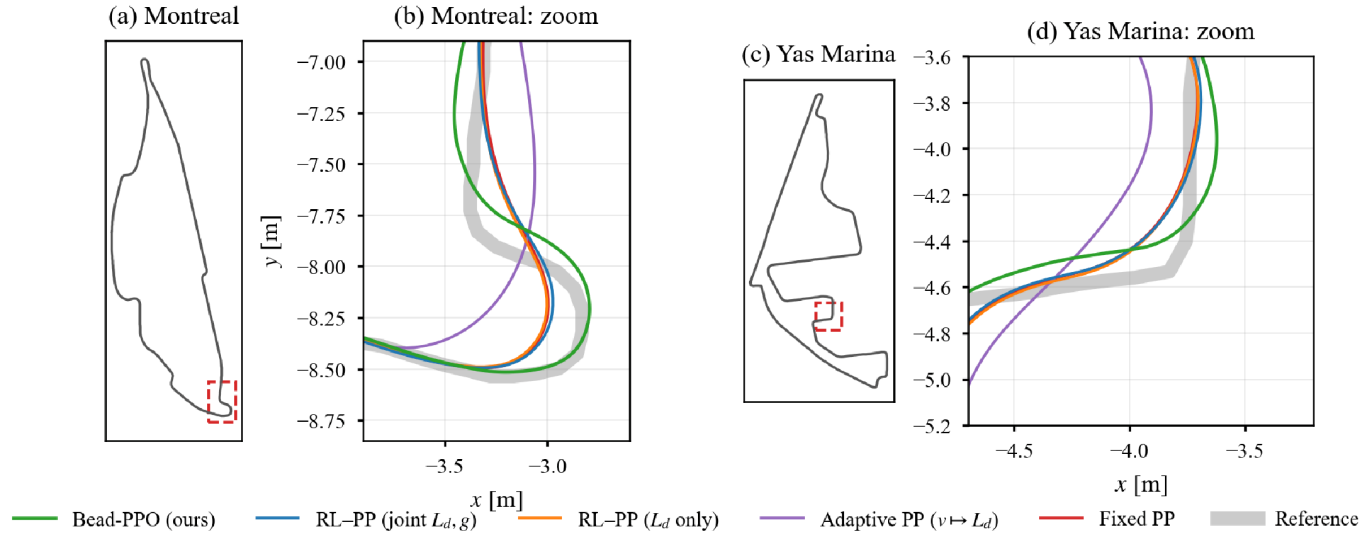


Fig. 7. Qualitative comparison at the most demanding corners. (a), (c) Full layouts with the zoom window marked. (b), (d) All five controllers overlaid on the reference.

F. Threats to Validity

Six limitations bound these conclusions. (i) *Training exposure is asymmetric:* Bead-PPO was trained on the three circuits it is scored on, whereas RL–PP saw none of them. The circuit comparison therefore favors our method, and Table IV is the fairer zero-shot test. (ii) *Model selection:* the DAgger stopping rule and checkpoint choice (Algorithm 1) use Geom@0.10 on the same three circuits, so the circuit results are not independent of model selection. (iii) *The dynamics differ:* the bead is holonomic and slower, whereas the PP agent obeys non-holonomic bicycle kinematics (Section V-B). (iv) *Actor-only training:* Bead-PPO uses the PPO parameterization, but its actor is shaped by supervised DAgger updates rather than by the clipped surrogate objective. It is thus closer to interactive imitation than to reward-driven learning and cannot exceed the expert on axes the expert ignores, such as speed. (v) *Single models:* each learned controller is one trained model, and the reported variability reflects only start-position perturbations, not training-seed variance. (vi) *Scale and hardware:* all results are in simulation at arena scale; no physical F1TENTH experiments were performed, and the RL–PP ranges and teacher were rescaled from [4].

VI. Conclusion

We asked whether a directly actuated policy tracks more precisely than a PP law with learned parameters when both face the same references and metrics, comparing Bead-PPO, a DAgger-trained point-mass tracker fed by an image-to-reference front end, with a same-arena re-implementation of RL-tuned PP [4] and two classical PP baselines.

On the three F1TENTH-derived circuits the answer is yes: Bead-PPO achieves the highest coverage ($\geq$95.7% at

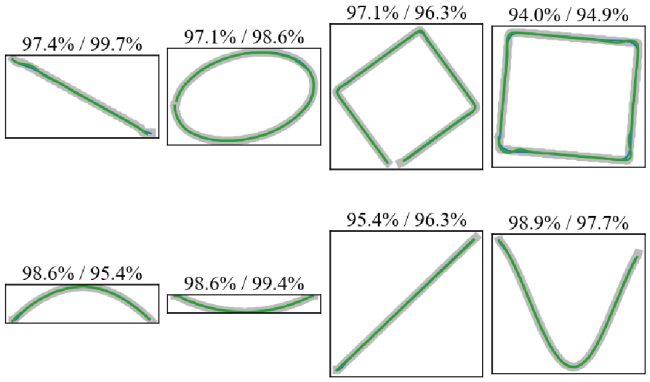


Fig. 8. A representative subset of the held-out closed shapes (top) and curves (bottom): Bead-PPO (green) and RL–PP joint (blue) on the reference (grey). Titles give Geom@0.10 as Bead-PPO / RL–PP.

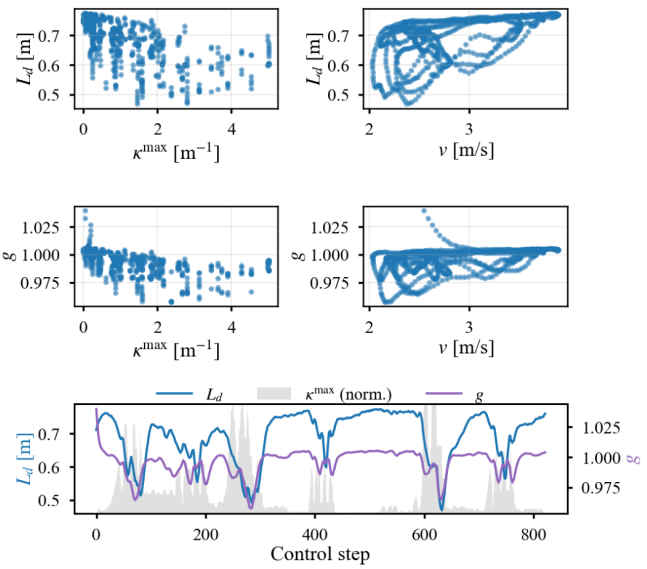


Fig. 9. Learned joint RL–PP schedule on Montreal. Top four panels: L_d and g versus previewed curvature $\kappa^{\max}$ and speed v. Bottom: along-lap evolution with normalized curvature shaded.

0.10 m) and the lowest RMSE, P95, and maximum error. That advantage holds at its native, damping-limited pace and on circuits also used for its training and model selection. On short held-out contours the answer is no: all methods perform comparably, and a hand-tuned PP is marginally better. Our re-implementation reproduces the qualitative lookahead behavior of RL-tuned PP but not its reported benefit from learning the steering gain.

Each of these caveats sets a direction for future work: (a) a common speed profile for both agents, to separate precision from pace; (b) held-out circuits for model selection and several training seeds; (c) PPO fine-tuning of the DAgger-initialized actor, so that reward can refine it beyond the expert; (d) a separately trained lookahead-only policy, for a faithful training-time ablation; and (e) transfer of the bead policy to a physical F1TENTH vehicle as a reference generator for a low-level steering loop.

Acknowledgment

The authors thank the F1TENTH community for the open racetrack data and Elgouhary and El-Wakeel [4] for a clearly specified formulation that made a same-arena re-implementation feasible.

References

[1] M. O'Kelly, H. Zheng, D. Karthik, and R. Mangharam, "F1TENTH: An open-source evaluation environment for continuous control and reinforcement learning," in *Proc. NeurIPS 2019 Competition and Demonstration Track*, PMLR vol. 123, 2020, pp. 77–89.

[2] R. C. Coulter, "Implementation of the pure pursuit path tracking algorithm," Carnegie Mellon Univ., Robotics Inst., Tech. Rep. CMU-RI-TR-92-01, 1992.

[3] J. M. Snider, "Automatic steering methods for autonomous automobile path tracking," Carnegie Mellon Univ., Robotics Inst., Tech. Rep. CMU-RI-TR-09-08, 2009.

[4] M. Elgouhary and A. S. El-Wakeel, "Learning to tune pure pursuit in autonomous racing: Joint lookahead and steering-gain control with PPO," arXiv:2602.18386, 2026.

[5] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[6] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. AISTATS*, 2011, pp. 627–635.

[7] G. M. Hoffmann, C. J. Tomlin, M. Montemerlo, and S. Thrun, "Autonomous automobile trajectory tracking for off-road driving: Controller design, experimental validation and racing," in *Proc. American Control Conf.*, 2007, pp. 2296–2301.

[8] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York, NY, USA: Springer, 2012.

[9] J. Kong, M. Pfeiffer, G. Schildbach, and F. Borrelli, "Kinematic and dynamic vehicle models for autonomous driving control design," in *Proc. IEEE Intelligent Vehicles Symp.*, 2015, pp. 1094–1099.

[10] A. Heilmeier, A. Wischnewski, L. Hermansdorfer, J. Betz, M. Lienkamp, and B. Lohmann, "Minimum curvature trajectory planning and control for an autonomous race car," *Vehicle System Dynamics*, vol. 58, no. 10, pp. 1497–1527, 2020.

[11] J. Betz *et al.*, "Autonomous vehicles on the edge: A survey on autonomous vehicle racing," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 458–488, 2022.

[12] B. R. Kiran *et al.*, "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, 2022.

[13] J. Kober, J. A. Bagnell, and J. Peters, "Reinforcement learning in robotics: A survey," *Int. J. Robotics Research*, vol. 32, no. 11, pp. 1238–1274, 2013.

[14] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Proc. ICLR*, 2016.

[15] M. Andrychowicz *et al.*, "What matters in on-policy reinforcement learning? A large-scale empirical study," in *Proc. ICLR*, 2021.

[16] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. ICML*, 2018, pp. 1861–1870.

[17] B. D. Evans, H. W. Jordaan, and H. A. Engelbrecht, "Comparing deep reinforcement learning architectures for autonomous racing," *Machine Learning with Applications*, vol. 14, p. 100496, 2023.

[18] D. A. Pomerleau, "ALVINN: An autonomous land vehicle in a neural network," in *Advances in Neural Information Processing Systems 1*, 1989, pp. 305–313.

[19] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, 1979.

[20] T. Y. Zhang and C. Y. Suen, "A fast parallel algorithm for thinning digital patterns," *Commun. ACM*, vol. 27, no. 3, pp. 236–239, 1984.

[21] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.

[22] M. Towers *et al.*, "Gymnasium: A standard interface for reinforcement learning environments," arXiv:2407.17032, 2024.